

Academic Stress & Student Well-being

An Independent Statistical Research Study

StudyBuild - Research Project 01

Sample	1,100 observations
Outcome	Stress level: Low / Medium / High
Main predictors	Study load, sleep quality, social support

Research Question

Are study load, sleep quality, and social support associated with students' stress levels? The data are observational, so the analysis is designed to estimate associations rather than causal effects.

Data

The project uses StressLevelDataset.csv from the Student Stress Factors: A Comprehensive Analysis dataset page on Kaggle. The CSV contains 1,100 observations and 21 variables. The analysis focuses on stress_level, study_load, sleep_quality, and social_support.

Data-quality checks found no missing values and no exact duplicate rows. Stress level contained only the expected codes 0, 1, and 2. The observed predictor ranges were 0-5 for study load, 0-5 for sleep quality, and 0-3 for social support.

Methods

The primary model was ordinal logistic regression because stress_level has three ordered categories. The predictors were entered as numeric ordered scores. This specification assumes that a one-unit increase has a comparable type of effect across each predictor scale, so coefficient interpretation is made cautiously.

- Focused cross-tabulations and percentage plots were used for exploratory analysis.
- Chi-square tests and Cramer's V summarized unadjusted categorical associations and effect size.
- Spearman correlations summarized the direction and strength of monotonic relationships.
- Variance inflation factors (VIF) were used to assess multicollinearity.
- Two threshold-specific binary logistic models were used as an exploratory proportional-odds diagnostic.
- Multinomial logistic regression was used as a sensitivity analysis when the proportional-odds assumption looked questionable.

Results

The stress groups were approximately balanced overall. The exploratory cross-tabulations showed strong but not perfectly monotonic relationships. For example, some score levels contained unusually concentrated stress categories, so the model results were interpreted together with assumption checks rather than as simple linear patterns.

Unadjusted associations

Predictor	Cramer's V	Spearman rho	Direction
Study load	0.602	0.666	Higher score, higher stress
Sleep quality	0.831	-0.761	Higher score, lower stress
Social support	0.688	-0.644	Higher score, lower stress

All three chi-square tests had $p < 0.001$, and all expected cell frequencies were above 5. Predictor VIF values were low (1.53-1.71), providing no evidence of serious multicollinearity among the three predictors.

Primary ordinal-logistic model

Predictor	Odds ratio	95% CI	p-value
Study load	2.16	1.85-2.51	< 0.001
Sleep quality	0.31	0.27-0.36	< 0.001
Social support	0.45	0.38-0.53	< 0.001

After adjustment for the other two predictors, a one-unit increase in study-load score was associated with about 2.16 times the odds of being in a higher stress category. Higher sleep-quality and social-support scores were associated with lower odds of being in a higher stress category. These are adjusted associations, not causal effects.

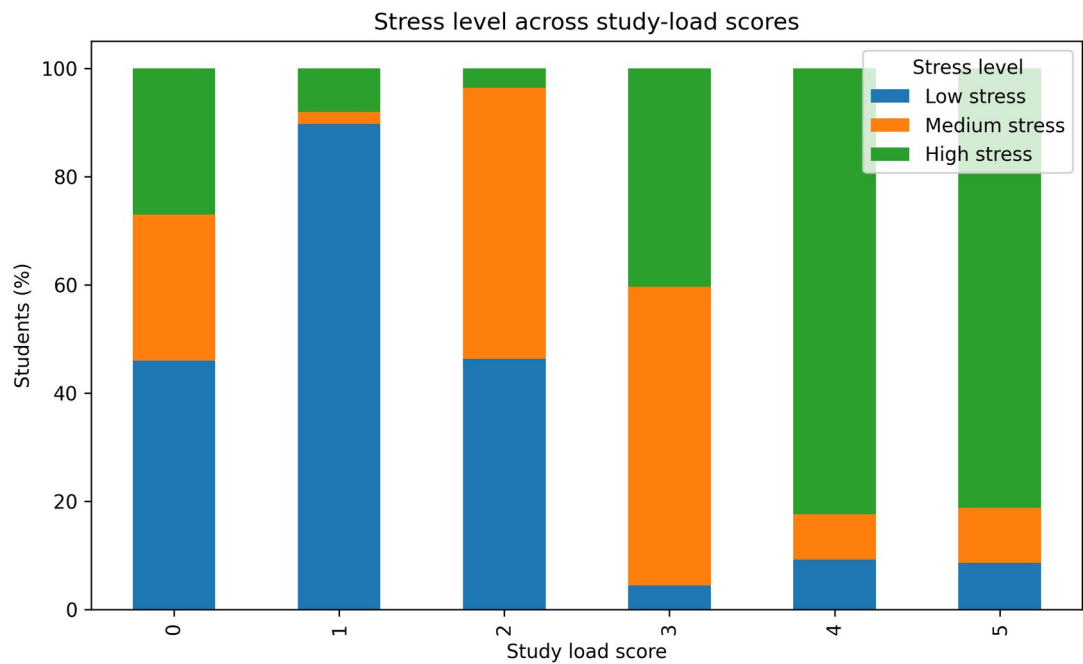


Figure 1. Observed stress distribution across study-load scores.

Assumption Check and Sensitivity Analysis

The proportional-odds assumption was explored by comparing predictor coefficients at two binary stress thresholds: Low versus Medium/High and Low/Medium versus High. Study-load coefficients were fairly similar (0.723 vs 0.638), sleep-quality coefficients differed somewhat (-1.151 vs -0.875), and social-support coefficients differed substantially (-0.323 vs -1.643). This made the proportional-odds assumption less convincing for social support.

Multinomial sensitivity analysis

Comparison	Predictor	OR	95% CI	p-value
Medium vs Low	Study load	1.89	1.54-2.32	< 0.001
Medium vs Low	Sleep quality	0.33	0.28-0.40	< 0.001
Medium vs Low	Social support	0.98	0.78-1.22	0.836
High vs Low	Study load	3.17	2.50-4.02	< 0.001
High vs Low	Sleep quality	0.18	0.14-0.22	< 0.001
High vs Low	Social support	0.17	0.12-0.24	< 0.001

Study load and sleep quality retained the same overall direction in the sensitivity analysis. Social support was different: after adjustment, it did not clearly distinguish medium from low stress, but it was strongly associated with high versus low stress. For this reason, the social-support result should not be summarized by a single proportional odds ratio without qualification.

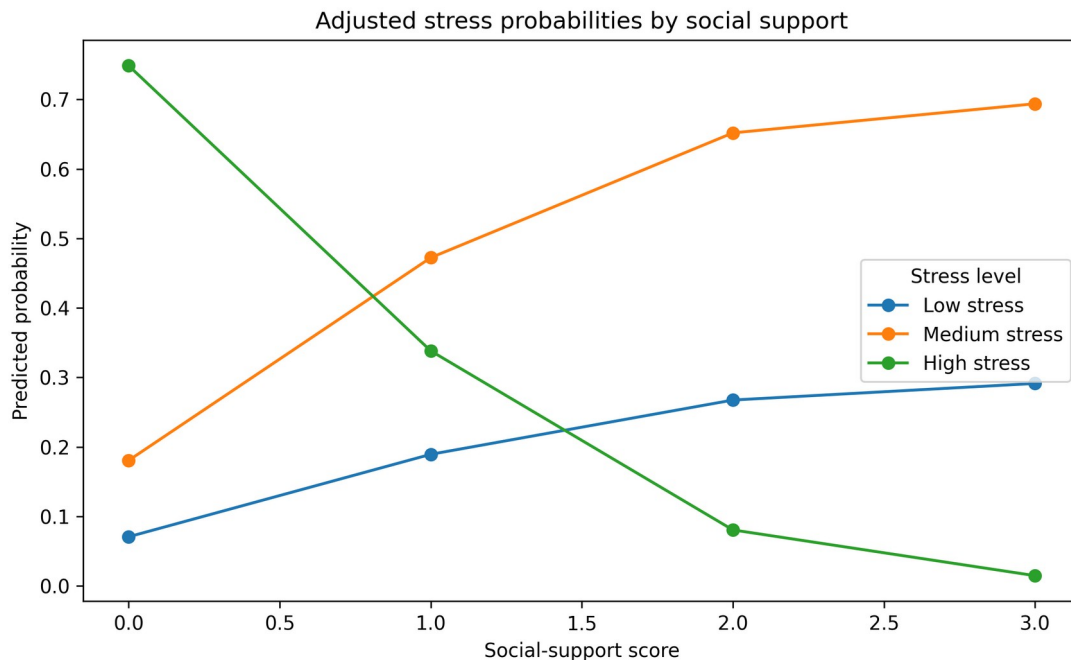


Figure 2. Adjusted stress probabilities across social-support scores from the multinomial model.

Discussion

The most consistent findings were for study load and sleep quality. Higher study-load scores were associated with higher stress, while better sleep-quality scores were associated with lower stress across both the ordinal and multinomial specifications. Social support was also related to stress, but the category-specific pattern in the sensitivity analysis shows that its relationship is more complex.

The analysis illustrates why model assumptions matter. The ordinal model is a natural first choice for an ordered outcome, but the threshold comparison suggested that a single proportional effect was not equally plausible for every predictor. The multinomial model was therefore used to check whether the substantive conclusions changed when that restriction was relaxed.

Limitations

- The dataset is observational, so causal conclusions are not supported.
- The full questionnaire wording and scale anchors for some integer-coded predictors are not available in the CSV, which limits substantive interpretation of one-unit changes.
- Information about the sampling frame is limited, so the findings should not automatically be generalized to all students.
- Other psychological, academic, and environmental factors may be related to stress and could confound the observed associations.
- Some relationships in the dataset are unusually strong, which should be considered when thinking about data generation and generalizability.
- The proportional-odds assumption was not equally convincing for all predictors, especially social support.

Conclusion

Study load, sleep quality, and social support were associated with stress level in this dataset. Higher study load was associated with higher stress, while better sleep quality was consistently associated with lower stress. Social support showed a less uniform pattern: it was strongly related to high versus low stress but did not clearly distinguish medium from low stress after adjustment in the multinomial model. The results should be described as associations rather than causes.

Reproducibility

The analysis is contained in notebooks/analysis.ipynb and runs from start to finish using the CSV in data/. Final figures are saved in figures/. Package versions are recorded at the end of the notebook. The CSV SHA-256 fingerprint is 14a45e92708b0c063ad4ab04563aa8fd4e3fc27157fd282e1c0658dc5161faed.

Dataset page: <https://www.kaggle.com/datasets/rxnach/student-stress-factors-a-comprehensive-analysis/data>